

Robotics Control System Simulation Instructions

Overview

This repository contains a Jupyter Notebook Template designed to assist participants in diagnosing and optimizing control code for robotic systems. It is tailored for Task 1 of the Johnson & Johnson Robotics & Controls Simulation.

The notebook includes exercises to:

- Diagnose response times for robotic arm commands.
- Identify and analyze delays in control code.
- Test and validate optimizations for improved performance.

Contents

- **Jupyter Notebook File:** Control System Diagnostics Template.ipynb
 - A structured notebook guiding users through diagnostic tasks.

Getting Started

1. **Download the Notebook:**
 - Click on the Control System Diagnostics Template.ipynb file.
 - Select "Download" to save it to your computer.
2. **Install Required Software (if not already installed):**
 - Install the Jupyter Notebook or JupyterLab. (Links for download are available below.)
 - Ensure Python 3.x is installed on your system. (Link for download are available below.)
3. **Open the Notebook:**
 - Open the .ipynb file in Jupyter Notebook, JupyterLab, or any compatible Python environment.
 - Launch Jupyter Notebook file by running the following command in your terminal or command prompt:

```
bash

jupyter notebook
```

4. Install Required Libraries:

- Ensure the following libraries are installed:

```
bash
```

```
pip install numpy
```

- Test the setup by running the following cell in notebook:

```
bash
```

```
import numpy as np
```

```
import time
```

```
print("Libraries are successfully installed!")
```

5. Follow the Instructions:

- The notebook includes step-by-step guidance to perform diagnostics, measure response times, and optimize control code.
- Execute each cell in sequence, following the prompts and instructions.

Requirements

- **Software:**
 - Jupyter Notebook, JupyterLab, or any Python 3.x-compatible environment (e.g., VS Code, Google Colab).
- **Libraries:**
 - *time*: Built into Python.
 - *numpy*: Install using the command provided above.

Troubleshooting Tips

1. **Error: *ModuleNotFoundError* for numpy:**
 - **Solution:** Run *pip install numpy* in your terminal or command prompt.
2. **Notebook Won't Open:**
 - **Solution:** Ensure Jupyter Notebook is installed and running. Check for installation issues.
3. **Kernel Not Found:**
 - **Solution:** Make sure Python 3.x is installed and selected as the kernel in Jupyter.
4. **File Not Executing:**
 - **Solution:** Ensure you are executing cells sequentially (top to bottom).